

Which kit do you have?

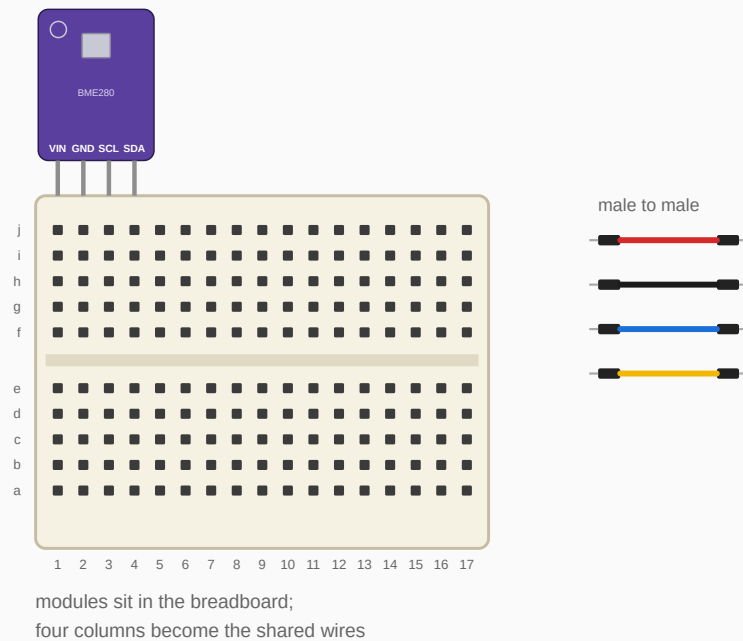
Two builds, one station. The difference is one part: a breadboard. Check the bag, pick a column, and follow only that kit's pages from here.

Kit A

Kit A: breadboard

A small plastic slab full of holes is in the bag.
Sensor and screen run at the same time, joined through the breadboard. Twelve jumper wires with a pin at both ends (male to male).

The US workshop kit and the UK bench.
About 25 minutes to a working station.



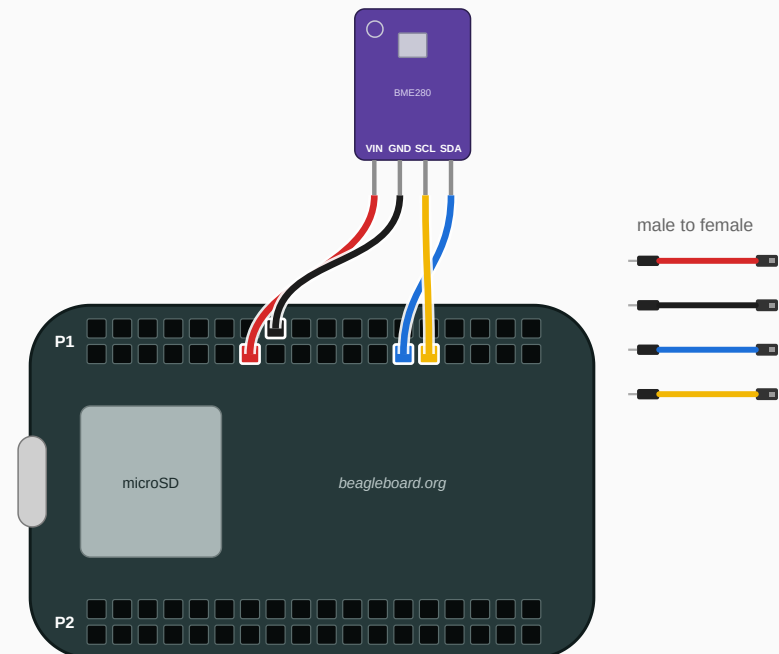
Kit B

Kit B: direct, no breadboard

No breadboard in the bag.
Four wires go straight from the board's sockets to one module. Sensor first; then the screen, on its own. The wires need a pin at one end and a socket at the other (male to female).

For any kit without a breadboard. Works anywhere. About 15 minutes.

one module at a time, wired straight from the board's sockets

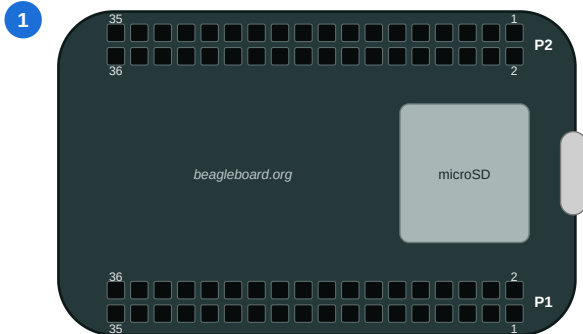


Same board, same four pins, same software, same checks. Only the joining differs. A Kit B bench becomes a Kit A bench the day a breadboard turns up.

Kit B: what is in the bag

Kit B: direct

No breadboard in this kit. Tip everything out and match each thing to a picture. Everything is drawn at the same scale as every later picture.



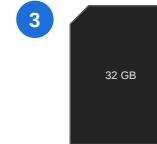
PocketBeagle 2

The computer. Two black strips of sockets (P1, P2) along the long edges, tiny numbers at the strip ends. Hold it by the edges. The side with the microSD slot and the dog logo is the side you use.



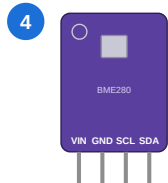
USB-C cable

Must be a data cable. A charge-only cable looks identical and is the most common reason nothing works.



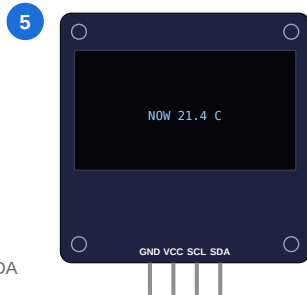
microSD card

8 GB or more. Holds the whole operating system.



BME280 sensor

Fingernail sized, purple or blue, four pins (a few have six): VIN or VCC, GND, SCL, SDA in tiny letters. The small metal can is the sensor itself. If the pins are loose in the bag instead of soldered on, stop: that is a different job.



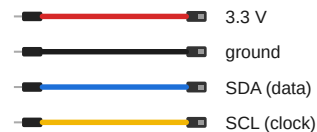
OLED screen (SSD1306, 0.96 inch)

A square board with a dark glass window and four pins on one edge: VCC, GND, SCL, SDA. The order of VCC and GND **differs between makers. Read yours now and write it down.** Getting those two backwards is the one mistake in this guide that kills a part instantly.

6 Jumper wires, male to female

A pin at one end, a socket at the other. Four needed. The pin end goes into the board's socket; the socket end slides over the module's pin.

Male-to-male wires will not reach the module.
Female-to-female wires will not reach the board.



No male-to-female wires? A male-to-male wire plus a female-to-female wire, joined end to end, does the same job.

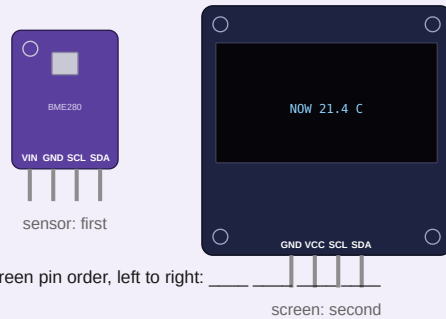
Not in the bag and not needed: a monitor, keyboard or power supply for the board. Your laptop is all three, through the USB-C cable.

Kit B: lay it out like this before you start

Kit B: direct

This is the arrangement every later picture uses. You sit at the bottom edge. Cable and card stay where they are until Parts 3 and 4.

1 Modules, pins toward you

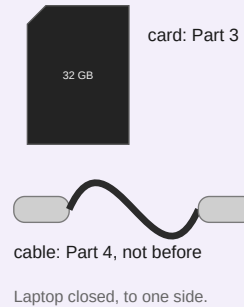


2 Four wires, one of each colour

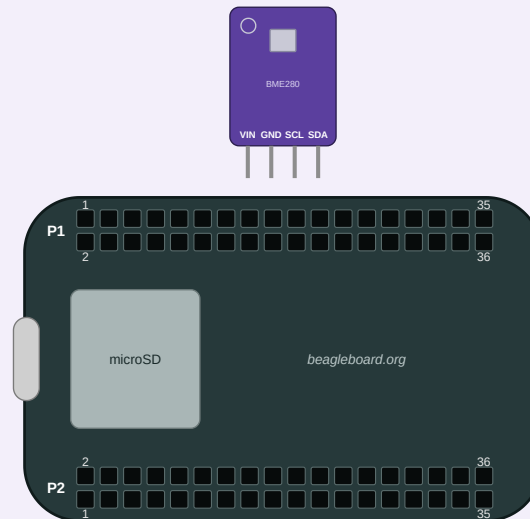


Pin end into the board's socket. Socket end over the module's pin.

3 Not yet



4 PocketBeagle 2, sockets up, USB-C port to the LEFT, the sensor 5 cm above it



P1 is the top strip, nearest the module.
Pin 1 is at the left end of P1, by the USB-C port.

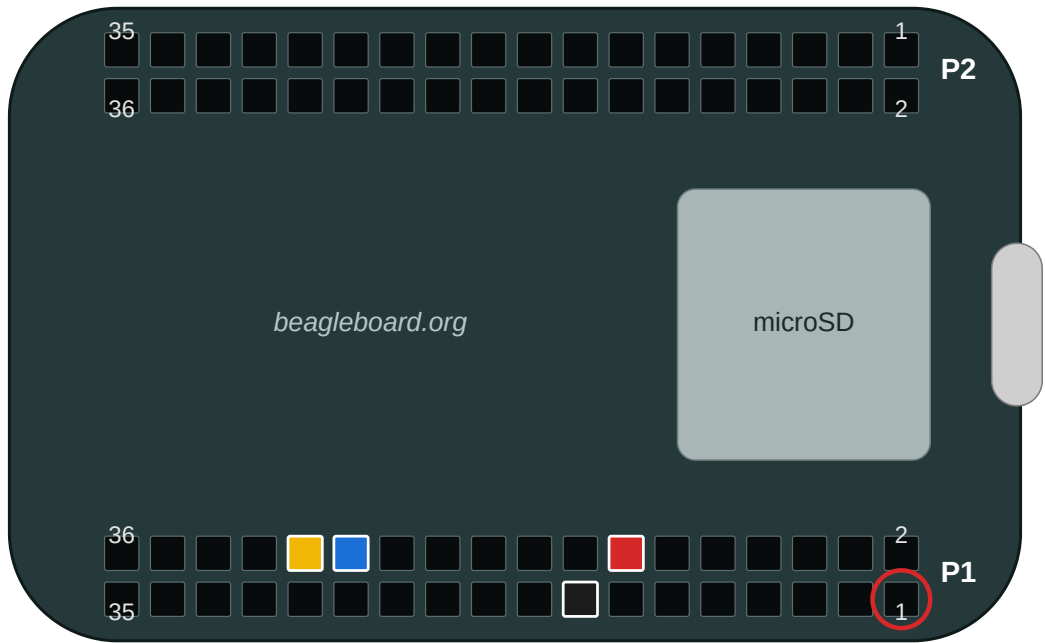
Find the tiny printed 1 and 2 there before going on.

The module just sits on the table above the board, pins toward it. Four short wires join them. Nothing to press into anything except the wire ends.

your table, seen from above

Where the four pins are

Board held with the sockets facing you and the USB-C port on the right, as BeagleBoard photographs it. On your table it is turned round; the numbers are the same.



P1.1 is 5 V. Nothing ever goes in here.

The four we use

- 14** **P1.14 VDD_3V3**
3.3 V out. column 7, inner row, from the USB-C end.
- 15** **P1.15 GND**
ground. column 8, outer row, from the USB-C end.
- 26** **P1.26 I2C2_SDA**
SDA, data. column 13, inner row, from the USB-C end.
- 28** **P1.28 I2C2_SCL**
SCL, clock. column 14, inner row, from the USB-C end.

Trust the numbers printed on the board over this picture. Find the printed 1 and 2 at the USB-C end of P1 and the 35 and 36 at the far end. If they are there, the columns are right.

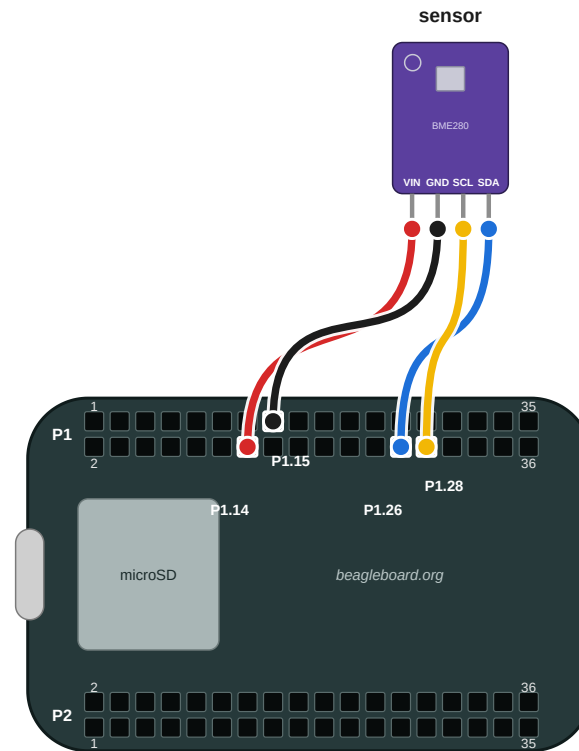
P1 close up. Column 1 is at the USB-C end. Outer row = odd pins, inner row = even pins.



Kit B, step 1 of 2: the sensor, wired straight to the board

Kit B: direct

Four male-to-female wires, pin end in the board, socket end on the module. Red P1.14 to VIN, black P1.15 to GND, blue P1.26 to SDA, yellow P1.28 to SCL. USB cable still in the bag.



- 3.3 V
- ground
- SDA (data)
- SCL (clock)

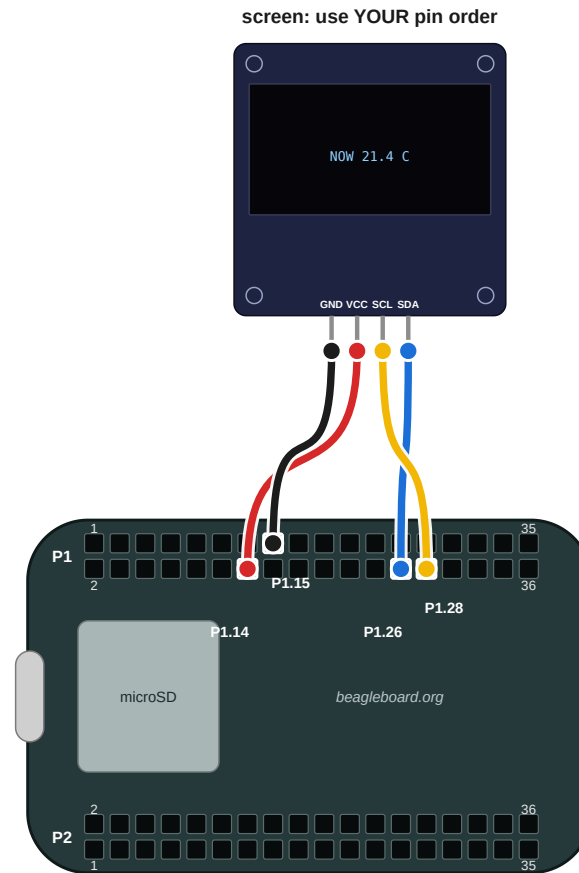
board turned so P1 faces the module: USB-C on the left, pin 1 at the left end of P1.

Say it out loud: red leaves 14 and arrives at VIN. Black leaves 15 and arrives at GND. Blue leaves 26 and arrives at SDA. Yellow leaves 28 and arrives at SCL. Nothing in P1.1. Then Part 3 (the card) and Part 4 (power). The sensor alone is a working station: readings come to your laptop with sws-live.

Kit B, step 2 of 2: the screen, on its own

Unplug the USB. Move the same four wires to the screen, in YOUR screen's pin order: red to VCC, black to GND, blue to SDA, yellow to SCL.

Kit B: direct



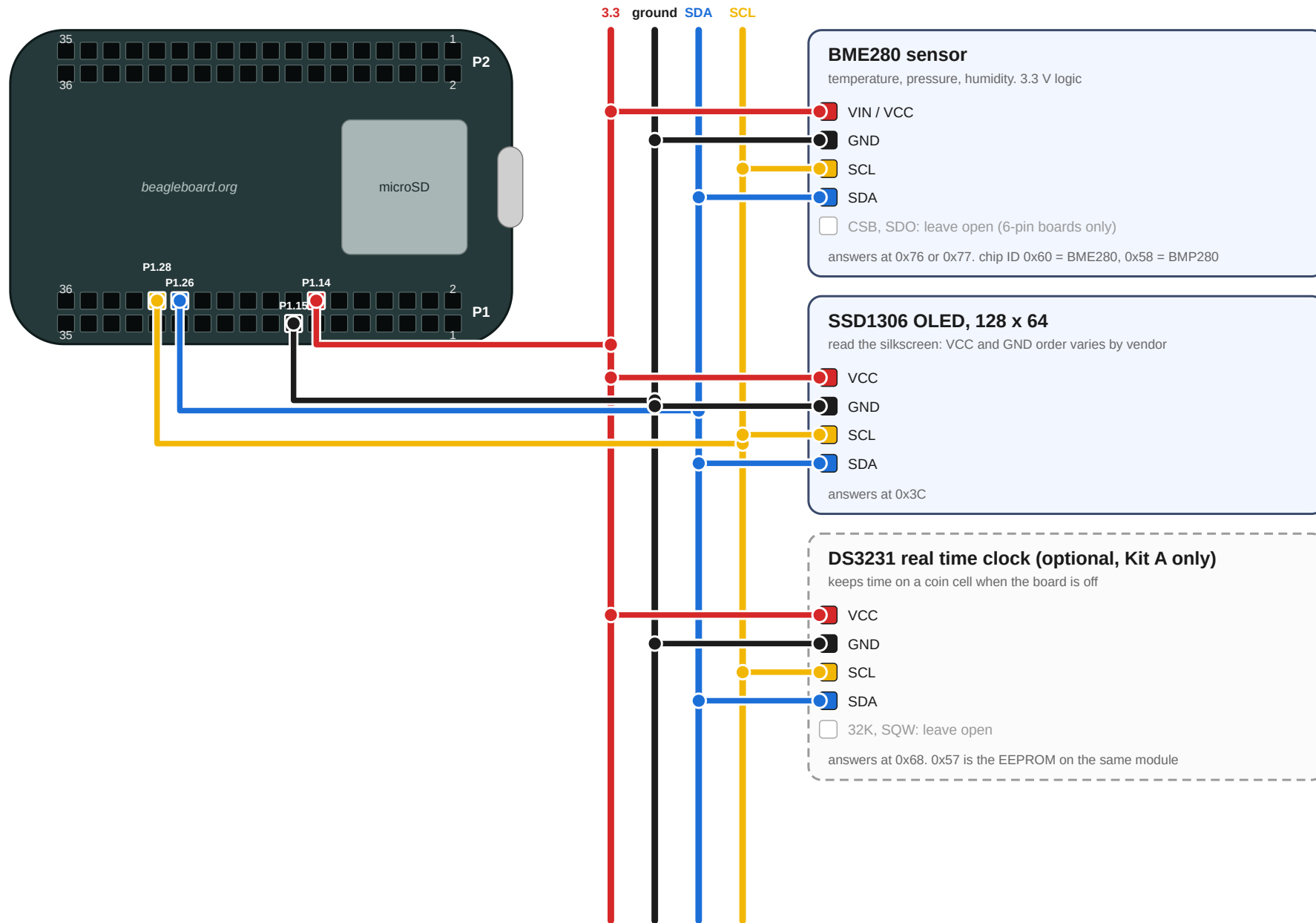
3.3 V
ground
SDA (data)
SCL (clock)

board turned so P1 faces the
module: USB-C on the left,
pin 1 at the left end of P1.

Only after the sensor has answered in Part 5. USB out first; never move a wire with power on. The screen alone shows the pages with -- where readings would be. A screen and a sensor together need a breadboard (Kit A). Until one arrives, this proves both parts and all four pins, which is most of the work.

The four wires, as a schematic

The same four wires for every module: power, ground, data, clock. This is the whole circuit, in either kit.



Kit A: four breadboard columns. Kit B: four wires, one module